$f: M \to M$ $\to$ surjection (injection) $\to$ f: iso

$\uparrow$

Noetherian (Artin~)

Artinian ring

$\Downarrow$ cc

$\mathbb{Z}$ $(2) \supset (2^2) \supset (2^3) \supset \cdots$

$M/\mathrm{Ker} f \cong M$

$\mathrm{Ker}\, T^n \cap \mathrm{Im}\, T^n$

$\mathrm{Ker}\, T^n$

$0 \to L \xrightarrow{f} M \xrightarrow{g} N \to 0$

$0 \to L \xrightarrow{f} M \Leftrightarrow f$: inj

$M \xrightarrow{g} N \to 0 \Leftrightarrow g$: sur

M: 노이터 $\Leftrightarrow$ L, N: 노이터

(Artin) (Artin)

$M/f(L) \cong N$

$M/L \cong N$

Cor M_i: 노이터 $(1 \le i \le n)$

$\Rightarrow \bigoplus_{i=1}^{n} M_i$ 노이터

$0 \to M_1 \to \bigoplus_{i=1}^{n} M_i \to \bigoplus_{i=1}^{n-1} M_i \to 0.$

$T: V \to V$ is diagonalizable $\Leftrightarrow$ m_T splits and separable

" triangularizable $\Leftrightarrow$ m_T is separable

relation {{

E/F is separable $\Leftrightarrow$ $\# \mathrm{Emb}_F(E, \bar{F}) = [E:F]$

E/F is splitting $\Leftrightarrow$ $\mathrm{Emb}_F(E, \bar{F}) = \mathrm{Aut}(E/F)$.